

2. Random Variables (Part I)

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Deep Generative Models: A Statistical Perspective



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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ This lecture briefly reviews random variables, univariate and multivariate distributions, and the expectations of their functions.
- ▶ Useful materials include:
 - ▶ Mathematics for Machine Learning [MML] (Deisenroth et al., 2020)
 - ▶ A First Course in Probability [FCP] (Ross, 2020)
 - ▶ Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- ▶ See Chapter 6 in [MML] for details. We present content mostly following the textbooks, but also provide substantial unique and advanced material.
- ▶ Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

1. Random Variables
2. Discrete and Continuous Probability Distributions
3. Summary Statistics and Independence
4. Multivariate Gaussian Distributions

Random Variables

- ▶ With the probability, we can describe probabilistic models of real-world phenomena involving randomness.
- ▶ In the two-coin toss example, suppose we are interested in the number of heads. Before the experiment, we do not know which $s \in S = \{HH, HT, TH, TT\}$ will occur, so the result is random (the probability describes this uncertainty).
- ▶ However, once the random experiment yields a specific s (e.g., HH), the number of heads is deterministically fixed (e.g., 2).
- ▶ **Random Variable**: A *random variable* (r.v.) is a function that maps each possible outcome in S to a quantity of interest in the *target space* $\mathcal{T}$.
- ▶ For example, $X : S = \{HH, HT, TH, TT\} \rightarrow \mathcal{T} = \{0, 1, 2\}$ defined by:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0 \quad (1)$$

is the r.v. representing the number of heads.

Random Variables

- ▶ Although the probability $P : \mathcal{A} \rightarrow [0, 1]$ and the r.v. $X : S \rightarrow \mathcal{T}$ have different domains, they are linked via the inverse image.
- ▶ For any subset $T \subseteq \mathcal{T}$, it is common to introduce the following notation:

$$P(X \in T) = P(\{s : X(s) \in T\}) = P(X^{-1}(T)). \quad (2)$$

- ▶ Formally, T must be a measurable set, meaning that $X^{-1}(T) \in \mathcal{A}$ so that its probability is defined under P .

Cumulative Distribution Function

- ▶ We first define distributions of univariate r.v.s ($X : S \rightarrow \mathbb{R}$), and then extend these concepts to multivariate ones (or *random vectors* $\mathbf{X} : S \rightarrow \mathbb{R}^d$).
- ▶ (Cumulative) Distribution Function: For a r.v. X ,

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty \quad (3)$$

is called the *cumulative distribution function (CDF)* or the *distribution function*.

- ▶ From now on, we omit the subscript denoting the random variable when the context is clear.
- ▶ In the two-coin toss example, where X represents the number of heads, the CDF of X is given by:

$$F(x) = \begin{cases} 0 & x < 0, \\ 1/4 & 0 \leq x < 1, \\ 3/4 & 1 \leq x < 2, \\ 1 & 2 \leq x. \end{cases} \quad (4)$$

Cumulative Distribution Function

► The CDF F satisfies the following properties:

1. F is a non-decreasing function.
2. $F(\infty) := \lim_{x \rightarrow \infty} F(x) = 1$.
3. $F(-\infty) := \lim_{x \rightarrow -\infty} F(x) = 0$.
4. F is right-continuous.

We denote the left limit at x by $F(x-)$, which satisfies $F(x-) = P(X < x)$.

► Conversely, for any function F satisfying the above four properties, there exists a probability P and a r.v. X such that $F_X = F$.

Discrete, Continuous, and Mixed Random Variables

► Types of Random Variables:

- X is *discrete* if F is a step function (piecewise constant).
- X is *continuous* if F is a continuous function.
- X is *mixed* if F is neither.

In this course, for continuous r.v.s, we assume that F is differentiable, except possibly at countably many points.

- The target space $\mathcal{T}$ is countable when X is discrete, and uncountable otherwise.

Discrete, Continuous, and Mixed Random Variables

- ▶ For a discrete r.v., it is convenient to define a function to describe the size of the jumps in its CDF.
- ▶ Probability Mass Function: For a discrete r.v. X , the function p_X defined by

$$p_X(x) := P(X = x) = P(\{s : X(s) = x\}) \quad (5)$$

is called the *probability mass function* (PMF).

- ▶ The PMF is related to the CDF by the jump size:

$$p(x) = P(X \leq x) - P(X < x) = F(x) - F(x-). \quad (6)$$

- ▶ For example, when X represents the number of heads in the two-coin toss,

$$p(1) = P(\{HT, TH\}) = 1/2. \quad (7)$$

Discrete, Continuous, and Mixed Random Variables

- ▶ Similarly, for a continuous r.v., it is convenient to define a function to describe the infinitesimal rate of increase of the CDF.
- ▶ **Probability Density Function**: For a continuous r.v. X , a function f_X satisfying the following conditions is called the *probability density function* (PDF):
 1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
 2. $P(a \leq X \leq b) = \int_a^b f_X(x) dx$ for any $a \leq b$.¹Note that $P(X = a) = \int_a^a f_X(x) dx = 0$ for any a .

¹This course uses f_X to distinguish density from probability, but it is also common to use p_X to denote the PDF.

Discrete, Continuous, and Mixed Random Variables

- ▶ The PDF is related to the CDF by its derivative:

$$f(x) = \frac{d}{dx}F(x) = \lim_{\epsilon \rightarrow 0} \frac{F(x + \epsilon) - F(x)}{\epsilon} \quad (8)$$

(defined at points where F is differentiable). Consequently, for a small $\epsilon > 0$,

$$P(x - \epsilon/2 \leq X \leq x + \epsilon/2) \approx f(x)\epsilon. \quad (9)$$

- ▶ The probability of an interval is given by the difference in CDF values:

$$P(a < X \leq b) = F(b) - F(a). \quad (10)$$

Means and Variances

- **Expectation (Mean):** The *expectation (mean)* of a discrete random variable X with probability mass function p_X is:

$$\mu_X = E[X] = \sum_x x p_X(x). \quad (11)$$

Correspondingly, the expectation of a continuous random variable X with probability density function f_X is:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx. \quad (12)$$

- **Law of the Unconscious Statistician:** If X is a discrete random variable, then for any real-valued function g , $E[g(X)] = \sum_x g(x) p_X(x)$. Similarly, for a continuous random variable X , $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$.²

²See [FCP] Proposition 4.1 for details.

Means and Variances

- ▶ **Variance:** The *variance* of X is defined as:

$$\sigma_X^2 = \text{Var}[X] = E[(X - E[X])^2]. \quad (13)$$

That is, $\text{Var}[X]$ is the expected squared Euclidean distance between the possible values of X and its mean.

- ▶ Important properties of means and variances include:
 - ▶ $E[aX + b] = aE[X] + b$
 - ▶ $\text{Var}[X] = E[X^2] - (E[X])^2$
 - ▶ $\text{Var}[aX + b] = a^2 \text{Var}[X]$
 - ▶ $\text{Var}[X] = 0$ if and only if X is deterministic, i.e., $P(X = E[X]) = 1$
- ▶ Using the mean and variance, the standardized random variable Z is defined as:

$$Z = \frac{X - E[X]}{\sqrt{\text{Var}[X]}} \quad (14)$$

where we assume $\text{Var}[X] > 0$.

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2. Discrete and Continuous Probability Distributions
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Univariate Discrete Random Variables

- ▶ Common univariate discrete distributions include the Bernoulli and binomial distributions.
- ▶ **Bernoulli Random Variable**: A random variable X is said to follow the *Bernoulli distribution*, denoted as $X \sim \text{Ber}(\mu)$ where $0 \leq \mu \leq 1$, if its PMF is given by:

$$p_X(x; \mu) = \begin{cases} \mu & \text{if } x = 1, \\ 1 - \mu & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases} \quad (15)$$

- ▶ The PMF of $X \sim \text{Ber}(\mu)$ can be summarized in the following table:

x	0	1
$p_X(x; \mu)$	$1 - \mu$	μ

- ▶ The mean and variance of X are $E[X] = \mu$ and $\text{Var}[X] = \mu(1 - \mu)$.

Univariate Discrete Random Variables

- **Binomial Random Variable:** A random variable X is said to follow the *binomial distribution*, denoted as $X \sim \text{Bin}(n, \mu)$, where $n \in \mathbb{N}$ and $0 \leq \mu \leq 1$, if its PMF is given by:

$$p_X(x; n, \mu) = \begin{cases} \binom{n}{x} \mu^x (1 - \mu)^{n-x} & \text{if } x \in \{0, 1, \dots, n\}, \\ 0 & \text{otherwise.} \end{cases} \quad (16)$$

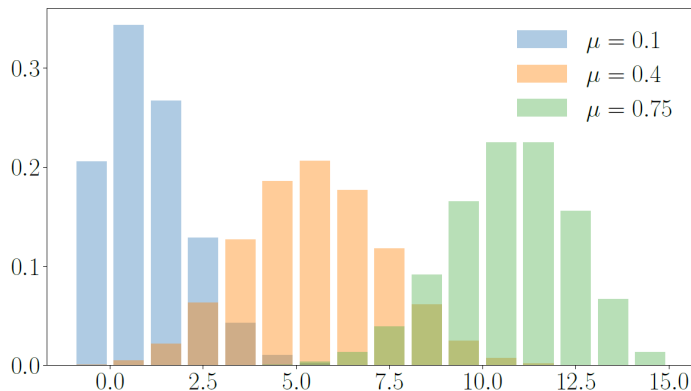
Here, $\binom{n}{x}$ denotes the binomial coefficient, defined as $\frac{n!}{x!(n-x)!}$, which represents the number of distinct ways to choose x objects from n distinguishable objects.

- The PMF of $X \sim \text{Bin}(n, \mu)$ can be summarized in the following table:

x	0	1	...	n
$p_X(x; n, \mu)$	$(1 - \mu)^n$	$n\mu(1 - \mu)^{n-1}$	...	μ^n

- The mean and variance of X are $E[X] = n\mu$ and $\text{Var}[X] = n\mu(1 - \mu)$.

Univariate Discrete Random Variables



[MML] Fig. 6.10: The PMFs of $\text{Bin}(n, \mu)$ when $(n, \mu) = (15, 0.1)$, $(15, 0.4)$, and $(15, 0.75)$.

Univariate Continuous Random Variables

- ▶ Common univariate continuous distributions include the uniform and Gaussian distributions.
- ▶ **Uniform Random Variable**: A random variable X is said to follow the *uniform distribution* over $[a, b]$, denoted by $X \sim \mathcal{U}(a, b)$, if its PDF is given by:

$$f_X(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases} \quad (17)$$

- ▶ The mean and variance of $X \sim \mathcal{U}(a, b)$ are given by:

$$\mathbb{E}[X] = \frac{a+b}{2} \quad \text{and} \quad \text{Var}[X] = \frac{(b-a)^2}{12}. \quad (18)$$

Univariate Continuous Random Variables

- ▶ The Gaussian distribution (or normal distribution) is undoubtedly one of the most fundamental and important distributions.
- ▶ The Gaussian distribution first appeared in 1738 during the analysis of the asymptotic behavior of binomial coefficients:

De Moivre-Laplace Theorem: Let $X \sim \text{Bin}(n, \mu)$. Then,

$$\lim_{n \rightarrow \infty} P \left(a \leq Z := \frac{X - E[X]}{\sqrt{\text{Var}[X]}} = \frac{X - n\mu}{\sqrt{n\mu(1-\mu)}} \leq b \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz. \quad (19)$$

- ▶ For a sufficiently large n , we obtain the following approximation:

$$\frac{X}{n} \approx \mu + \sqrt{\frac{\mu(1-\mu)}{n}} Z, \quad (20)$$

where Z has the PDF $\phi(z) := \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$.

Univariate Continuous Random Variables

- ▶ This motivates the definition of a class of random variables of the form $\mu + \sigma Z$, where Z is a random variable with the probability density function (PDF):

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \quad (21)$$

- ▶ These are called Gaussian distributions, and Z is specifically referred to as the *standard Gaussian random variable*.
- ▶ **Gaussian Random Variable**: A random variable X is said to follow the *Gaussian distribution (or normal distribution)* with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$, if:

$$X = \mu + \sigma Z, \quad (22)$$

for some standard Gaussian random variable $Z \sim \mathcal{N}(0, 1)$.

Univariate Continuous Random Variables

- When $X \sim \mathcal{N}(\mu, \sigma^2)$, its PDF is given by:

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (23)$$

This is a popular alternative definition of the Gaussian distribution.

Proof:

$$\begin{aligned} f_X(x; \mu, \sigma^2) &= \frac{d}{dx} P(X \leq x) = \frac{d}{dx} P\left(Z \leq \frac{x - \mu}{\sigma}\right) \\ &= \frac{d}{dx} \int_{-\infty}^{(x - \mu)/\sigma} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\ &= \frac{1}{\sigma} \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right) \right). \end{aligned} \quad (24)$$

Univariate Continuous Random Variables

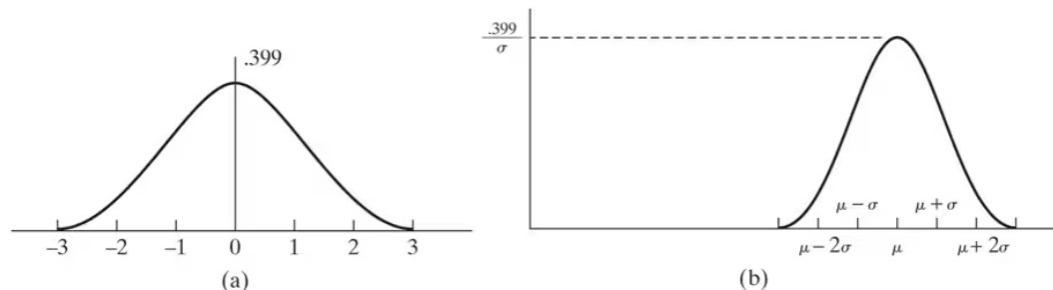


Fig. 5.5 [FCP]: Normal density function: (a) $\mu = 0, \sigma = 1$; (b) arbitrary μ, σ^2 .

- We denote the PDF and CDF of the *standard Gaussian distribution* $\mathcal{N}(0, 1)$ by $\phi(z)$ and $\Phi(z)$, respectively.

Multivariate Random Variables

- ▶ Joint (Cumulative) Distribution Function: For a random vector $\mathbf{X} := (X_1, \dots, X_d)^T$, its joint CDF is given by:

$$F_{\mathbf{X}}(\mathbf{x}) = P(X_1 \leq x_1, \dots, X_d \leq x_d) \quad (25)$$

where $\mathbf{x} := (x_1, \dots, x_d)^T \in \mathbb{R}^d$.

- ▶ Joint Probability Mass Function: For a jointly discrete random vector $\mathbf{X}$, its joint PMF $p_{\mathbf{X}}$ is given by:

$$p_{\mathbf{X}}(\mathbf{x}) := P(\mathbf{X} = \mathbf{x}) = P(\{s : X_1(s) = x_1, X_2(s) = x_2, \dots, X_d(s) = x_d\}). \quad (26)$$

- ▶ The joint PMF is also related to the joint CDF. For example, when $d = 2$,

$$p(x_1, x_2) = F(x_1, x_2) - F(x_1-, x_2) - F(x_1, x_2-) + F(x_1-, x_2-). \quad (27)$$

Multivariate Random Variables

- Example 6.1a from [FCP]: Suppose that three balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. Let X and Y denote the number of red and white balls chosen, respectively. Then, the joint PMF of X and Y is given by:

$$p_{X,Y}(i,j) = \frac{\binom{3}{i} \binom{4}{j} \binom{5}{3-i-j}}{\binom{12}{3}} \quad (28)$$

for non-negative integers i and j such that $i + j \leq 3$.

Multivariate Random Variables

$i \backslash j$					
	0	1	2	3	Row sum = $P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum = $P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

[FCP] Tab. 6.1: $p_{X,Y}(i,j) = P(X = i, Y = j)$.

Multivariate Random Variables

- **Multinomial Random Vector**: A random vector $\mathbf{X}$ is said to follow the *multinomial distribution*, denoted by $\mathbf{X} \sim \text{Multin}(n, \boldsymbol{\mu})$, where $n \in \mathbb{N}$, $\boldsymbol{\mu} = (\mu_1, \dots, \mu_d)^T$, $\mu_i \geq 0$, and $\sum_{i=1}^d \mu_i = 1$, if its PMF is given by:

$$p(\mathbf{x}; n, \boldsymbol{\mu}) = \begin{cases} \binom{n}{x_1, x_2, \dots, x_d} \mu_1^{x_1} \mu_2^{x_2} \cdots \mu_d^{x_d} & \text{if } x_i \in \{0, 1, \dots, n\} \text{ and } \sum_{i=1}^d x_i = n, \\ 0 & \text{otherwise.} \end{cases} \quad (29)$$

- When $\mathbf{X} \sim \text{Multin}(n, \boldsymbol{\mu})$, the marginal distribution of the i -th coordinate is binomial:

$$X_i \sim \text{Bin}(n, \mu_i). \quad (30)$$

Multivariate Random Variables

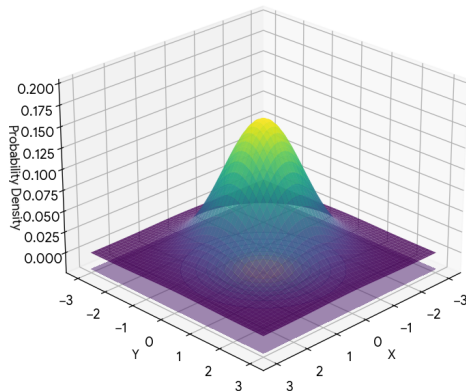
- ▶ Joint Probability Density Function: For a jointly continuous random vector $\mathbf{X}$, its joint PDF $f_{\mathbf{X}}$ satisfies:
 1. $f_{\mathbf{X}}(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^d$.
 2. $P(\mathbf{X} \in (a_1, b_1) \times \cdots \times (a_d, b_d)) = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f_{\mathbf{X}}(x_1, \dots, x_d) dx_d \dots dx_1$.
- ▶ For example, let $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ be the PDFs of X_1 and X_2 , respectively. Then, the product

$$f_{X_1}(x_1)f_{X_2}(x_2), \tag{31}$$

which is a function of $(x_1, x_2)^T \in \mathbb{R}^2$, satisfies the conditions for a joint PDF.

Multivariate Random Variables

Standard Bivariate Normal PDF
(Volume under surface = Probability)



- When f_{X_1} and f_{X_2} are the PDFs of $\mathcal{N}(0, 1^2)$,

$$\begin{aligned}(x_1, x_2)^T &\mapsto f_{X_1}(x_1)f_{X_2}(x_2) \\ &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}(x_1^2 + x_2^2)\right)\end{aligned}\quad (32)$$

defines a joint PDF of $\mathbf{X} = (X_1, X_2)^T$.

Multivariate Random Variables

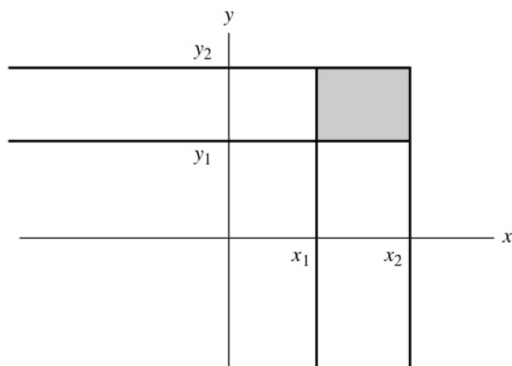
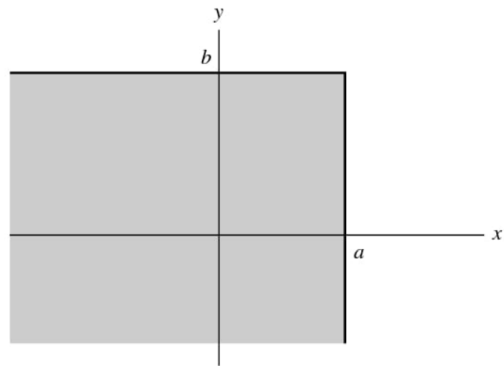
- ▶ The joint PDF is related to the joint CDF by its derivative:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^d}{\partial x_1 \cdots \partial x_d} F_{\mathbf{X}}(\mathbf{x}). \quad (33)$$

Consequently, for a small $\epsilon_i > 0$,

$$P(x_1 - \epsilon_1/2 \leq X_1 \leq x_1 + \epsilon_1/2, \dots, x_d - \epsilon_d/2 \leq X_d \leq x_d + \epsilon_d/2) \approx f_{\mathbf{X}}(\mathbf{x}) \epsilon_1 \cdots \epsilon_d. \quad (34)$$

Multivariate Random Variables



Figs. 3.1 and 3.2 [MSDA]:

$$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1)$$

Multivariate Random Variables

- Given $p_{X,Y}(x, y)$, we can calculate:

$$P(X = x) = P(X = x, Y \in \mathcal{Y}) = \sum_{y \in \mathcal{Y}} P(X = x, Y = y) = \sum_{y \in \mathcal{Y}} p_{X,Y}(x, y). \quad (35)$$

- **Marginal Probability Mass Function:** Let $\mathbf{X}$ and $\mathbf{Y}$ be random vectors with joint PMF $p_{\mathbf{X},\mathbf{Y}}(\mathbf{x}, \mathbf{y})$. Then, the *marginal PMF* $p_{\mathbf{X}}(\mathbf{x})$ is given by:

$$p_{\mathbf{X}}(\mathbf{x}) = \sum_{\mathbf{y} \in \mathcal{Y}} p_{\mathbf{X},\mathbf{Y}}(\mathbf{x}, \mathbf{y}) \quad (36)$$

Multivariate Random Variables

- Given the joint PDF $f_{X,Y}(x,y)$, we can derive the marginal PDF as follows:

$$\begin{aligned} f_X(x) &:= \lim_{\epsilon \rightarrow 0^+} \frac{P(x - \epsilon/2 \leq X \leq x + \epsilon/2)}{\epsilon} = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\epsilon} \int_{-\infty}^{\infty} \int_{x-\epsilon/2}^{x+\epsilon/2} f_{X,Y}(u,y) du dy \\ &= \int_{-\infty}^{\infty} \left(\lim_{\epsilon \rightarrow 0^+} \frac{1}{\epsilon} \int_{x-\epsilon/2}^{x+\epsilon/2} f_{X,Y}(u,y) du \right) dy \\ &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy. \end{aligned} \tag{37}$$

- Marginal Probability Density Function: Let $\mathbf{X}$ and $\mathbf{Y}$ be random vectors with joint PDF $f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$. Then, the *marginal PDF* of $\mathbf{X}$, denoted $f_{\mathbf{X}}(\mathbf{x})$, is given by:

$$f_{\mathbf{X}}(\mathbf{x}) = \int_{\mathbb{R}^{d_Y}} f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) d\mathbf{y}. \tag{38}$$

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Means and Covariances

- For a multivariate r.v. $\mathbf{X}(\cdot) := (X_1(\cdot), \dots, X_d(\cdot))^T$, its expectation is defined as:

$$\mu_{\mathbf{X}} = \mathbb{E}_{\mathbf{X}}[\mathbf{X}] = (\mathbb{E}_{X_1}[X_1], \dots, \mathbb{E}_{X_d}[X_d])^T \in \mathbb{R}^d. \quad (39)$$

- The expectation operator is linear. For any real constants a and b , and any real-valued functions g and h defined on the support of $\mathbf{X}$,

$$\mathbb{E}[ag(\mathbf{X}) + bh(\mathbf{X})] = a \mathbb{E}[g(\mathbf{X})] + b \mathbb{E}[h(\mathbf{X})]. \quad (40)$$

Means and Covariances

- ▶ Covariance: The *covariance* between two univariate r.v.s X and Y is given by:

$$\sigma_{X,Y} = \text{Cov}_{X,Y}[X, Y] = E_{X,Y} [(X - E_X[X])(Y - E_Y[Y])]. \quad (41)$$

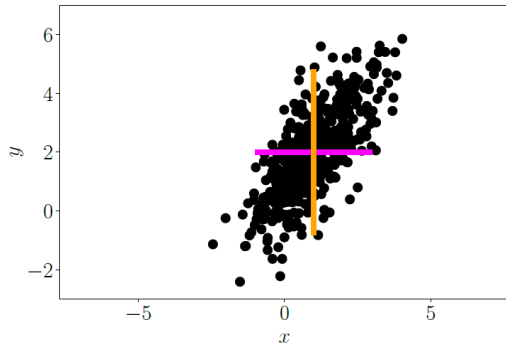
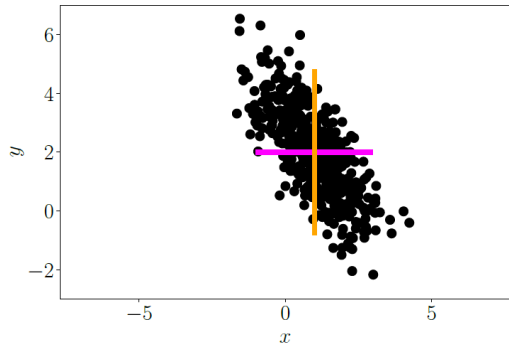
- ▶ For a d_X -dimensional $\mathbf{X}$ and a d_Y -dimensional $\mathbf{Y}$, their covariance is defined as:

$$\Sigma_{\mathbf{X},\mathbf{Y}} = \text{Cov}_{\mathbf{X},\mathbf{Y}}[\mathbf{X}, \mathbf{Y}] = E_{\mathbf{X},\mathbf{Y}}[(\mathbf{X} - E_{\mathbf{X}}[\mathbf{X}])(\mathbf{Y} - E_{\mathbf{Y}}[\mathbf{Y}])^T] \in \mathbb{R}^{d_X \times d_Y}. \quad (42)$$

- ▶ A useful shortcut formula is:

$$\text{Cov}[\mathbf{X}, \mathbf{Y}] = E[\mathbf{X}\mathbf{Y}^T] - E[\mathbf{X}]E[\mathbf{Y}]^T. \quad (43)$$

Means and Covariances



[MML] Fig. 6.5: Scatter plots of two observed random variables X and Y .

Means and Covariances

- Variance: The *variance* of a r.v. $\mathbf{X}$ is defined as:

$$\Sigma_{\mathbf{X}} = \text{Var}_{\mathbf{X}}[\mathbf{X}] = \mathbb{E}_{\mathbf{X}}[(\mathbf{X} - \mathbb{E}_{\mathbf{X}}[\mathbf{X}])(\mathbf{X} - \mathbb{E}_{\mathbf{X}}[\mathbf{X}])^T] \quad (44)$$

This matrix is referred to as the *covariance matrix* of $\mathbf{X}$.

- Since $\text{Var}[\mathbf{X}] = \text{Cov}[\mathbf{X}, \mathbf{X}]$, based on the shortcut formula of covariance,

$$\text{Var}[\mathbf{X}] = \mathbb{E}[\mathbf{X}\mathbf{X}^T] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}^T]. \quad (45)$$

- For any functions g and h defined on the support of $\mathbf{X}$ and $\mathbf{Y}$, respectively,

$$\text{Cov}[g(\mathbf{X}), h(\mathbf{Y})] = \mathbb{E} \left[(g(\mathbf{X}) - \mathbb{E}[g(\mathbf{X})]) (h(\mathbf{Y}) - \mathbb{E}[h(\mathbf{Y})])^T \right]. \quad (46)$$

Means and Covariances

- ▶ **Correlation:** Let X and Y be univariate random variables having positive variances. The *correlation* between X and Y is given by:

$$\rho_{X,Y} = \text{Corr}_{X,Y}[X, Y] = \frac{\text{Cov}_{X,Y}[X, Y]}{\sqrt{\text{Var}_X[X] \text{Var}_Y[Y]}}. \quad (47)$$

- ▶ By the Cauchy-Schwartz inequality, $-1 \leq \text{Corr}(X, Y) \leq 1$ always holds.
- ▶ Again, the *standardized variable* (or *Z-score*) of X is defined as:

$$Z_X = \frac{X - E[X]}{\sqrt{\text{Var}[X]}}. \quad (48)$$

The correlation is equal to the covariance between Z -scores:

$$\text{Corr}[X, Y] = \text{Cov} \left[\frac{X - E[X]}{\sqrt{\text{Var}[X]}}, \frac{Y - E[Y]}{\sqrt{\text{Var}[Y]}} \right]. \quad (49)$$

Means and Covariances

- ▶ **Sample Mean (Empirical Mean):** Let $x_1, \dots, x_n$ be n observations of a random variable X . Then, the *sample mean* of X is defined as $\bar{x} = n^{-1} \sum_{i=1}^n x_i$.
- ▶ **Sample Covariance:** Let $(x_1, y_1)^T, \dots, (x_n, y_n)^T$ be n observations of a random vector $(X, Y)^T$. Then, the *sample covariance* between X and Y is defined as $s_{X,Y} = (n-1)^{-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$.
- ▶ **Sample Variance:** Let $x_1, \dots, x_n$ be n observations of a random variable X . Then, the *sample variance* of X is defined as $s_{X,X} = s_X^2 = (n-1)^{-1} \sum_{i=1}^n (x_i - \bar{x})^2$.
- ▶ **Sample Correlation:** Let $(x_1, y_1)^T, \dots, (x_n, y_n)^T$ be n observations of a random vector $(X, Y)^T$. Assume $s_X > 0$ and $s_Y > 0$. Then, the *sample correlation* is defined as $r_{X,Y} = s_{X,Y} / (s_X s_Y)$.
- ▶ We can similarly define multivariate versions of the sample mean and covariance matrix.

Means and Covariances

- ▶ For any two random vectors $\mathbf{X} \in \mathbb{R}^d$ and $\mathbf{Y} \in \mathbb{R}^d$:
 - ▶ $E[\mathbf{X} + \mathbf{Y}] = E[\mathbf{X}] + E[\mathbf{Y}]$.
 - ▶ $E[\mathbf{X} - \mathbf{Y}] = E[\mathbf{X}] - E[\mathbf{Y}]$.
 - ▶ $\text{Var}[\mathbf{X} + \mathbf{Y}] = \text{Var}[\mathbf{X}] + \text{Var}[\mathbf{Y}] + \text{Cov}[\mathbf{X}, \mathbf{Y}] + \text{Cov}[\mathbf{X}, \mathbf{Y}]^T$.
 - ▶ $\text{Var}[\mathbf{X} - \mathbf{Y}] = \text{Var}[\mathbf{X}] + \text{Var}[\mathbf{Y}] - \text{Cov}[\mathbf{X}, \mathbf{Y}] - \text{Cov}[\mathbf{X}, \mathbf{Y}]^T$.
- ▶ Let $\mathbf{X} \in \mathbb{R}^{d_X}$ and $\mathbf{Y} \in \mathbb{R}^{d_Y}$ be random vectors. For any constant matrices A and B , and constant vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ of appropriate dimensions:
 - ▶ $E[A\mathbf{X} + \mathbf{b}] = A E[\mathbf{X}] + \mathbf{b}$.
 - ▶ $\text{Var}[A\mathbf{X} + \mathbf{b}] = A \text{Var}[\mathbf{X}] A^T$.
 - ▶ $\text{Cov}[A\mathbf{X} + \mathbf{c}, B\mathbf{Y} + \mathbf{d}] = A \text{Cov}[\mathbf{X}, \mathbf{Y}] B^T$.

Statistical Independence

- ▶ **Statistical Independence:** Two random variables X and Y are said to be *independent* if, for any two (measurable) sets T_X and T_Y in their target spaces,

$$P(X \in T_X, Y \in T_Y) = P(X \in T_X)P(Y \in T_Y). \quad (50)$$

That is, the events $X^{-1}(T_X)$ and $Y^{-1}(T_Y)$ are independent.

- ▶ We write $X \perp\!\!\!\perp Y$ to denote that X is independent of Y .
- ▶ When the target spaces are real coordinate spaces, two random vectors $\mathbf{X}$ and $\mathbf{Y}$ are independent if and only if either of the following equivalent conditions holds:
 1. $F_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = F_{\mathbf{X}}(\mathbf{x})F_{\mathbf{Y}}(\mathbf{y})$
 2. $f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = f_{\mathbf{X}}(\mathbf{x})f_{\mathbf{Y}}(\mathbf{y})$ (or $p_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = p_{\mathbf{X}}(\mathbf{x})p_{\mathbf{Y}}(\mathbf{y})$ in the discrete case).
- ▶ For any (measurable) functions g and h , if $X \perp\!\!\!\perp Y$, then $g(X) \perp\!\!\!\perp h(Y)$.

Statistical Independence

- ▶ The random vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are said to be independent if, for all measurable sets $T_1, T_2, \dots, T_n$ in their respective target spaces,

$$P(\mathbf{X}_1 \in T_1, \mathbf{X}_2 \in T_2, \dots, \mathbf{X}_n \in T_n) = \prod_{i=1}^n P(\mathbf{X}_i \in T_i). \quad (51)$$

- ▶ When the target spaces are real coordinate spaces, these random vectors are independent if and only if the following equivalent conditions hold:
 1. $F_{\mathbf{X}_1, \dots, \mathbf{X}_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = F_{\mathbf{X}_1}(\mathbf{x}_1) \cdots F_{\mathbf{X}_n}(\mathbf{x}_n)$
 2. $f_{\mathbf{X}_1, \dots, \mathbf{X}_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = f_{\mathbf{X}_1}(\mathbf{x}_1) \cdots f_{\mathbf{X}_n}(\mathbf{x}_n)$ (or $p_{\mathbf{X}_1, \dots, \mathbf{X}_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = p_{\mathbf{X}_1}(\mathbf{x}_1) \cdots p_{\mathbf{X}_n}(\mathbf{x}_n)$ in the discrete case).
- ▶ An infinite collection of random vectors is said to be independent if every finite sub-collection is independent.

Statistical Independence

- ▶ For any two independent random vectors $\mathbf{X} \in \mathbb{R}^d$ and $\mathbf{Y} \in \mathbb{R}^d$:

$$\text{Cov}[\mathbf{X}, \mathbf{Y}] = 0, \quad (52)$$

implying $\text{Var}[\mathbf{X} + \mathbf{Y}] = \text{Var}[\mathbf{X}] + \text{Var}[\mathbf{Y}]$. However, the converse does not hold in general.

- ▶ **Example 6.5 from [MML]:** Consider a random variable X with $E[X] = 0$ and $E[X^3] = 0$. Let $Y = X^2$. Then, $\text{Cov}[X, Y] = 0$. Do you think Y is independent of X ? That is, if we observe X , does this observation truly provide no information regarding Y ?

Outline

1. Random Variables
2. Discrete and Continuous Probability Distributions
3. Summary Statistics and Independence
4. Multivariate Gaussian Distributions

Multivariate Gaussian Distributions

- ▶ We first define the *multivariate standard Gaussian random vector* $\mathbf{Z}$:

$$f_{\mathbf{Z}}(\mathbf{z}) = \prod_{i=1}^d \phi(z_i) = (2\pi)^{-d/2} \exp\left(-\mathbf{z}^T \mathbf{z} / 2\right), \quad \text{denoted by } \mathbf{Z} \sim \mathcal{N}(0, I_d). \quad (53)$$

- ▶ The multivariate Central Limit Theorem states that the sample mean vector is approximately distributed as:

$$\frac{\mathbf{X}_1 + \cdots + \mathbf{X}_n}{n} \approx \boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z}, \quad (54)$$

for some $\boldsymbol{\mu} \in \mathbb{R}^d$, $\Sigma^{1/2} \in \mathbb{R}^{d \times d}$, and $\mathbf{Z} \sim \mathcal{N}(0, I_d)$, where $\Sigma^{1/2}$ is symmetric and $\Sigma^{1/2} \Sigma^{1/2} := \Sigma$ is symmetric and positive definite.

Multivariate Gaussian Distributions

- ▶ **Multivariate Gaussian Distribution**: A random vector $\mathbf{X} \in \mathbb{R}^d$ is said to follow the *multivariate Gaussian distribution* with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and symmetric and positive definite matrix $\Sigma \in \mathbb{R}^{d \times d}$, denoted by $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if:

$$\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z}, \quad \text{where } \mathbf{Z} \sim \mathcal{N}(0, I_d). \quad (55)$$

We also use the notation $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma)$ to refer to the multivariate Gaussian PDF.

- ▶ To derive the joint PDF of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, we first derive the change of variables formula, which describes how a probability density transforms under a one-to-one differentiable mapping.

Multivariate Gaussian Distributions

- Let $\mathbf{X}$ be a continuous random vector with PDF $f_{\mathbf{X}}(\mathbf{x})$ supported on a region $T_{\mathbf{X}} \subseteq \mathbb{R}^d$. Let $\mathbf{Y} = g(\mathbf{X})$, where g is a one-to-one, differentiable transformation from $T_{\mathbf{X}}$ onto $T_{\mathbf{Y}}$. Then, $\mathbf{Y}$ has the PDF:

$$f_{\mathbf{Y}}(\mathbf{y}) = \begin{cases} f_{\mathbf{X}}(g^{-1}(\mathbf{y})) \left| \det \frac{\partial g^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right|, & \text{if } \mathbf{y} \in T_{\mathbf{Y}}, \\ 0, & \text{otherwise.} \end{cases} \quad (56)$$

Here, $\left| \det \frac{\partial g^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right|$ denotes the absolute value of the determinant of the Jacobian matrix.

Multivariate Gaussian Distributions

- When $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is differentiable and invertible on T_X , and $g^{-1}(\mathbf{y}) \in T_X$, denoting the hypercube $C_\epsilon(\mathbf{y}) = [\mathbf{y} - \frac{\epsilon}{2}\mathbf{1}, \mathbf{y} + \frac{\epsilon}{2}\mathbf{1}]$ where $\mathbf{1} := (1, \dots, 1)^T$:

$$\begin{aligned}
 f_Y(\mathbf{y}) &= \lim_{\epsilon \rightarrow 0^+} \frac{P(\mathbf{Y} \in C_\epsilon(\mathbf{y}))}{\epsilon^d} \\
 &= \lim_{\epsilon \rightarrow 0^+} \frac{P(\mathbf{X} \in g^{-1}(C_\epsilon(\mathbf{y})))}{\epsilon^d} \\
 &= \lim_{\epsilon \rightarrow 0^+} \frac{P(\mathbf{X} \in g^{-1}(C_\epsilon(\mathbf{y})))}{\text{Vol}(g^{-1}(C_\epsilon(\mathbf{y})))} \frac{\text{Vol}(g^{-1}(C_\epsilon(\mathbf{y})))}{\epsilon^d} \\
 &= f_X(g^{-1}(\mathbf{y})) \left| \det \frac{\partial g^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right|.
 \end{aligned} \tag{57}$$

Multivariate Gaussian Distributions

- **Change of Variables Formula:** Suppose there exists a partition of T_X such that $T_X = \bigcup_{i=1}^k T_{X,i}$ (where $T_{X,i} \cap T_{X,j} = \emptyset$ for $i \neq j$), and the restriction of g to $T_{X,i}$, denoted by g_i , is a one-to-one differentiable mapping. Then, $\mathbf{Y}$ has the PDF:

$$f_{\mathbf{Y}}(\mathbf{y}) = \sum_{i=1}^k f_{\mathbf{X}}(g_i^{-1}(\mathbf{y})) \left| \det \frac{\partial g_i^{-1}(\mathbf{y})}{\partial \mathbf{y}} \right|, \quad (58)$$

where the sum is over indices i such that $\mathbf{y}$ is in the image of g_i .

- The change of variables formula is fundamental in defining multivariate distributions.

Multivariate Gaussian Distributions

- Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. That is, $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2}\mathbf{Z}$ where $\mathbf{Z} \sim \mathcal{N}(0, I_d)$. Then, the PDF of $\mathbf{X}$ is given by:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right). \quad (59)$$

Sketch of Proof: Note that $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ implies that $g(\mathbf{z}) = \boldsymbol{\mu} + \Sigma^{1/2}\mathbf{z}$ is a one-to-one differentiable mapping. By the change of variables formula,

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = f_{\mathbf{Z}}(g^{-1}(\mathbf{x})) \left| \det \frac{\partial g^{-1}(\mathbf{x})}{\partial \mathbf{x}} \right|. \quad (60)$$

Multivariate Gaussian Distributions

- For example, for any bivariate Gaussian random vector

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right), \quad (61)$$

its PDF is given by:

$$f_{X,Y}(x,y;\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\left(\frac{x-\mu_X}{\sigma_X} \right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) \right) \right]. \quad (62)$$

Multivariate Gaussian Distributions

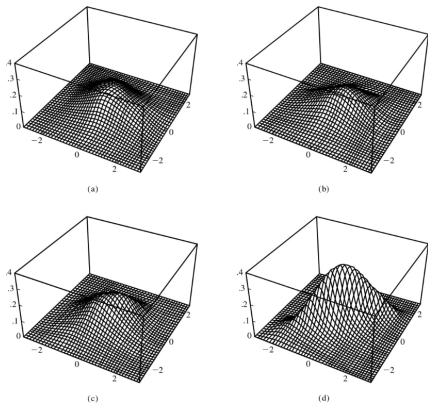


Fig. 3.9. [MSDA]

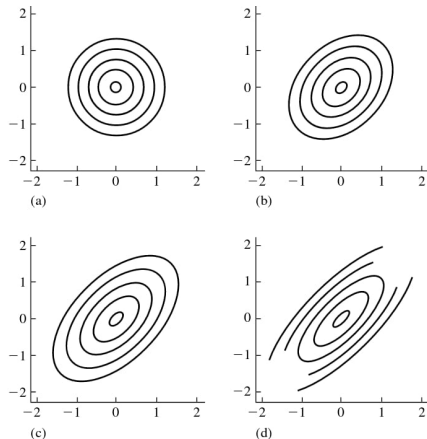


Fig. 3.10. [MSDA]

References I

- Deisenroth, M. P., Faisal, A. A., and Ong, C. S. (2020). *Mathematics for machine learning*. Cambridge University Press.
- Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.
- Ross, S. M. (2020). *A first course in probability*. Pearson Harlow, UK.